Antonin LECOCQ

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Third-year Mathematics Student at Université Paris Dauphine – PSL combining research experience in astrophysics and mathematics. Comfortable moving between rigorous theory, implementation and application.

Education

- **Université Paris Dauphine – PSL** Sep. 2024 – Present
  Bachelor's Degree in Applied Mathematics *Paris, France*
  - **Coursework** Calculus, Algebra, Probability Theory, Algorithms and Programming, English...
- **EIB Étoile High School** Sep. 2021 – Jun. 2024
  French General Baccalaureate with high honors *Paris, France*
  - **Specializations** Mathematics, Physics and Chemistry, and Computer Science.

Experience

- **Astrophysics Research Intern – UROP PSL** Nov. 2026 – Present
  Observatoire de Paris – PSL *Paris, France*
  - Analysis of the Praesepe open cluster using Gaia DR4 data, membership determination.
- **Mathematics Research Intern** Jun. 2026 – Jul. 2026
  MAP5, Université Paris Cité *Paris, France*
  - Study of the $SO(n)$ Lie group, reviewed English-language literature, and wrote a research report. Developed geometric registration models using Python and PyTorch.
- **Backend Developer Intern** Jul. 2025
  Celesta *Lisbon, Portugal*
  - Developed Symfony APIs within a microservices architecture, refactored PHP code, and worked with Docker. Designed and implemented new KYC functionality.
- **Treasurer & Webmaster** Oct. 2024 – May 2026
  Bureau des Arts de Dauphine *Paris, France*
  - **Treasurer**: Financial management of artistic projects, fundraising and event organization.
  - **Webmaster**: Security, maintenance, and updates of three websites.
- **Private Math Tutoring** *(Oct. 2026 – Present)* – Paris, France
- **Sea Rescuer** *(Aug. 2022)* – Cavalaire-sur-Mer Tourist Office, France
- **Waiter** *(Jul. 2021)* – Taokan, Paris, France

Research

Lecocq, Antonin (2026). *Study of the Lie group SO(n) in the context of shape analysis*. Tech. rep. Under the supervision of Rayane Mouhli. Université Paris Cité, MAP5. DOI: 10.5281/zenodo.21557282.

Technical Skills

Programming	Python, C, PHP, Haskell, Bash Unix, R, LaTeX
Tools & Libraries	Git, Unix/Linux, Microsoft Excel, NumPy, Matplotlib, PyTorch
Languages	French, English, Italian

Certifications & Interests

GQS Trainer & PSC First Aid – French Red Cross	May 2025
Level 2 Diver – FFESSM	Aug. 2023
Recreational Boat License, Coastal Option – French Ministry of the Sea	May 2023
Aeronautical Initiation Certificate (BIA) – Ministry of National Education	Jun. 2022
Advanced Open Water Diver – PADI	Mar. 2022

First Place, at the Globeducate France Robotics Challenge 2023
Ranked #6868 among 100,000+ users on Root-Me, with 1,980 points
Interests: Digital music composition, hiking, and scuba diving.